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Sexual size dimorphism and multivariate phenotypic selection in a flightless weevil --Manuscript Draft--

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Corresponding Author:	Clint D. Kelly University of Quebec in Montreal Montreal, PQ CANADA
First Author:	Clint D. Kelly
Order of Authors:	Clint D. Kelly
Abstract:	Sexual size dimorphism (SSD) can arise from sex differences in fecundity, sexual and viability selection, but contemporary fitness measures do not necessarily reveal the processes that produced dimorphism. I examined morphology, egg load, and pairing status in an undescribed flightless New Zealand weevil (<i>Lyperobates</i> sp. A), focusing on body and hind-leg length. Female body length was positively associated with the number of mature eggs present at dissection, consistent with fecundity selection favouring larger females, although egg load does not measure lifetime reproductive output. Directional selection gradients based on pairing status were qualitatively similar in both sexes: body length was positively associated with pairing status, whereas hind-leg length was negatively associated with it. Thus, females and males did not experience opposing directional selection regimes. The male analysis supported nonlinear and correlational selection, with the highest predicted pairing probability among males combining relatively large bodies with short legs. However, a pooled analysis provided little evidence that the body length $\times$ leg length association differed between the sexes, so this contrast should not be interpreted as male-specific correlational selection. Females were larger than males in both traits, and body length was not assortative among the 25 observed pairs. Because pairing status was measured during a single sampling period, it may reflect encounter, mounting, or attachment duration rather than lifetime mating or reproductive success. These results document female-biased SSD and similar contemporary directional pairing-status selection in both sexes, but do not establish that the measured selection components caused or currently maintain the dimorphism.
Opposed Reviewers:	



Département des sciences biologiques
Université du Québec à Montréal
Montreal, Québec, Canada

T (514) 987-3000 ext. 0374
F (514) 987-4647
kelly.clint@uqam.ca

Dear Editors,

Please consider my manuscript, "Sexual size dimorphism and multivariate phenotypic selection in a flightless weevil," for publication as a Research Paper in *Animal Behaviour*.

Understanding how selection acts on multiple, correlated traits is essential for explaining the evolution of sexual dimorphism. However, multivariate selection is rarely estimated in both sexes within the same natural population, particularly in species for which mating success can be assessed directly in the field. In this study, I examined sexual size dimorphism and phenotypic selection on body and leg length in a flightless New Zealand weevil, *Lyperobates* sp. A. I compared individuals observed in mating pairs with unpaired males and females collected from the same population.

Females were larger than males in both body and leg length, but multivariate analyses revealed broadly similar directional selection in the two sexes: mating success increased with body length but declined with leg length after accounting for their positive correlation. The male analysis additionally provided evidence of nonlinear and correlational selection, suggesting that male mating success depended on particular combinations of body and leg dimensions. Importantly, a pooled analysis provided little evidence that the directional or correlational selection gradients differed statistically between the sexes. My results therefore illustrate how similar multivariate selection can act on sexually dimorphic traits and highlight the importance of distinguishing apparent sex-specific patterns from statistically supported differences between the sexes.

I believe the manuscript is well suited to *Animal Behaviour* because it addresses the evolution of mating success, sexual dimorphism and sex-specific selection using field observations of naturally interacting animals. The study also contributes data on an endemic, flightless invertebrate and demonstrates the value of jointly analysing correlated morphological traits rather than interpreting separate univariate associations.

This manuscript is original, has not been published previously and is not under consideration elsewhere. I have disclosed funding sources in the manuscript. The research complied with applicable requirements governing the collection and treatment of wildlife in New Zealand and followed the Association for the Study of Animal Behaviour/Animal Behavior Society Guidelines for the Treatment of Animals in Behavioural Research and Teaching.

Thank you for considering my manuscript. I hope that it will be of interest to readers of *Animal Behaviour*.

Sincerely,

A handwritten signature in black ink, appearing to read 'Clint Kelly', is written over a light blue horizontal line.

Clint D. Kelly
Département des sciences biologiques
Université du Québec à Montréal
Montréal, Québec, Canada

Sexual size dimorphism and multivariate phenotypic selection in a flightless weevil

Clint D. Kelly^{1,✉}

¹ Département des Sciences biologiques, Université du Québec à Montréal, Montréal, Canada

✉Correspondence: [Clint D. Kelly <kelly.clint@uqam.ca>](mailto:kelly.clint@uqam.ca)

Data availability

Data are available through the OSF repository at

https://osf.io/khqgs/overview?view_only=728733529d06410ab17086f648a49a38.

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Conflict of Interest

I declare no conflict of interest.

Acknowledgements

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1 **Sexual size dimorphism and multivariate phenotypic**
2 **selection in a flightless weevil**

3

Abstract

Sexual size dimorphism (SSD) can arise from sex differences in fecundity, sexual and viability selection, but contemporary fitness measures do not necessarily reveal the processes that produced dimorphism. I examined morphology, egg load, and pairing status in an undescribed flightless New Zealand weevil (*Lyperobates* sp. A), focusing on body and hind-leg length. Female body length was positively associated with the number of mature eggs present at dissection, consistent with fecundity selection favouring larger females, although egg load does not measure lifetime reproductive output. Directional selection gradients based on pairing status were qualitatively similar in both sexes: body length was positively associated with pairing status, whereas hind-leg length was negatively associated with it. Thus, females and males did not experience opposing directional selection regimes. The male analysis supported nonlinear and correlational selection, with the highest predicted pairing probability among males combining relatively large bodies with short legs. However, a pooled analysis provided little evidence that the body length $\times$ leg length association differed between the sexes, so this contrast should not be interpreted as male-specific correlational selection. Females were larger than males in both traits, and body length was not assortative among the 25 observed pairs. Because pairing status was measured during a single sampling period, it may reflect encounter, mounting, or attachment duration rather than lifetime mating or reproductive success. These results document female-biased SSD and similar contemporary directional pairing-status selection in both sexes, but do not establish that the measured selection components caused or currently maintain the dimorphism.

Introduction

Sexual size dimorphism (SSD) is a widespread and evolutionarily significant feature of animal morphology (Darwin, 1871; Fairbairn et al., 2007). Its direction and magnitude can reflect sex-specific fecundity and sexual selection, as well as viability selection, ecological divergence, developmental processes, and genetic constraints or correlations between the sexes (Blanckenhorn, 2000; Fairbairn, 1997; Fairbairn et al., 2007). In many species, male-biased SSD is associated with sexual selection arising from male–male competition or female choice (Andersson, 1994), whereas female-biased SSD is often associated with fecundity selection because larger females can produce or carry more eggs (Blanckenhorn, 2005; Honek, 1993; Kingsolver & Pfennig, 2004; Pincheira-Donoso & Hunt, 2017; Zhu et al., 2025). Nevertheless, a positive relationship between female body length and fecundity does not, on its own, demonstrate that fecundity selection produced SSD. Dimorphism is an evolved outcome of selection acting across generations, whereas estimates from a single population and time period describe contemporary phenotypic associations with particular components of fitness.

When sexual selection favours increased male body length, as often occurs under male–male contest competition, male-biased SSD may evolve, or female-biased SSD may be reduced (Andersson, 1994; Emlen & Oring, 1977). Conversely, female-biased SSD has sometimes been attributed to weak sexual selection on males or to selection favouring smaller, more mobile males (Fairbairn et al., 2007; Ghiselin, 1974). Scramble-competition polygyny is frequently invoked in this context because male reproductive success can depend on locating and securing females rather than defending resources (Andersson, 1994; Moya-Laraño et al., 2002). Smaller males, or males with locomotor morphology that facilitates mate searching, have higher mating success in some systems, including Japanese beetles (*Popillia japonica*) and Cook Strait giant

48 weta (*Deinacrida rugosa*) (Kelly et al., 2008; Kelly, 2020; Kelly & Gwynne, 2023).
49 Comparative evidence, however, shows that larger males are also favoured in many scramble-
50 competitive arthropods (Herberstein et al., 2017). The direction of selection on male morphology
51 therefore cannot be inferred from a broad mating-system label alone, especially when mate
52 searching and pair formation have not been observed directly.

53 Multivariate analyses are important because the association between one trait and a fitness
54 component can depend on other traits (Brodie et al., 1995; Lande & Arnold, 1983; Phillips &
55 Arnold, 1989). For example, body length may influence competitive ability or persistence,
56 whereas leg length may influence movement or the ability to remain attached to a female.
57 Directional gradients describe the average slope of the phenotype–fitness relationship, while
58 quadratic and correlational gradients describe the curvature and dependence between traits.
59 These latter components may reveal combinations of morphology associated with high fitness
60 even when the directional gradients for individual traits appear similar between groups.

61 Assortative mating provides a related but distinct perspective on how morphology structures
62 mating interactions. Positive assortment occurs when individuals with similar trait values pair
63 more often than expected, whereas negative assortment occurs when dissimilar individuals pair
64 preferentially (Crespi, 1989; Jiang et al., 2013). Assortment can influence the covariance
65 between male and female phenotypes, alter the distribution of mating success, and thereby affect
66 the opportunity for sexual selection and the evolutionary dynamics of sexually dimorphic traits
67 (McDonald & Pizzari, 2016). An absence of assortment does not imply an absence of selection
68 on morphology: a trait may affect whether an individual is paired without determining its
69 partner’s phenotype.

Weevils (Curculionoidea) encompass substantial ecological and mating-system diversity across approximately 62,000 species (Haran et al., 2023), including around 1,500 species endemic to New Zealand (May, 1993; see also Brown, 2017). *Lyperobates* are flightless, ground-dwelling New Zealand weevils typically associated with alpine and subalpine habitats (Figure 1). The study population was nocturnally active; individuals were commonly found on kawakawa (*Piper excelsum*), and males were sometimes observed mounted on females for extended periods. I did not observe pair formation, systematic male–male contests, or female choice; consequently, the apparent absence of resource defence is consistent with, but does not demonstrate, a scramble-competitive mating system. Here, I examine SSD and contemporary phenotypic selection in an undescribed species of *Lyperobates* (Coleoptera: Curculionidae: Entiminae; hereafter *Lyperobates* sp. A).

I addressed four questions. First, is female body length positively associated with egg load, as expected under fecundity selection? Second, how are body length, hind leg length, and their combination associated with the probability that females and males were observed paired, and is there evidence that the correlational association differs between the sexes? Third, are females and males dimorphic in body and hind leg length, and do the contemporary selection gradients align with the direction and magnitude of that dimorphism? Finally, is body length assortative within observed mating pairs?

Methods

Study species and field sampling

I studied a population of an undescribed *Lyperobates* weevil (hereafter *Lyperobates* sp. A) on Te Pākeka/Maud Island, New Zealand. This taxon is currently under formal description but can be

distinguished from described congeners by consistent differences in body length, colouration, and elytral morphology (C.D. Kelly, personal observation). All individuals used in this study conformed to the same diagnostic phenotype and are treated as a single operational taxonomic unit.

Adult weevils were collected on 11 nights between 20 April and 5 May 2007. Individuals were located opportunistically by searching the ground and low vegetation (≤ 2 m) along a 20-m section of track bordering forest habitat. Sampling began approximately 1 h after sunset (approximately 18:30 h) and continued for 2 h. Weevils were found almost exclusively on kawakawa leaves. I collected each observed male–female pair ($N = 25$), together with singleton males ($N = 75$) and females ($N = 52$), yielding 100 males and 77 females in total. Because collection was opportunistic and paired individuals may differ from singletons in detectability, this sample ratio should not be interpreted as an estimate of the population or operational sex ratio. Each individual was placed in a separate 50-mL Falcon tube, assigned a unique identification code, transported to a field laboratory on Maud Island, and subsequently euthanized by freezing.

All individuals were photographed alongside a ruler for scale. Females ($N = 69$) were dissected, and egg load was quantified as the number of eggs present in the reproductive tract at the time of collection. Egg load provides an index of current reproductive investment, but it may be influenced by female age, mating history, oviposition history, and the timing of egg maturation; it is therefore not equivalent to lifetime realized fecundity. Morphological measurements were obtained from digital images using Fiji (Schindelin et al., 2012). Body length was measured from the anterior edge of the thorax to the posterior edge of the abdomen, and hind leg length was

114 measured on the third leg pair from the proximal end of the femur to the distal end of the tibia, to
115 the nearest 0.01 mm.

116 **Ethical note**

117 This study involved behavioural observations and morphological measurements of adult
118 *Lyperobates* sp. A weevils. Efforts to minimize stress and disturbance included minimizing
119 handling and maintaining individuals under appropriate conditions during the study. Following
120 completion of the behavioural observations, weevils were humanely euthanized by freezing
121 before dissection and photography. Formal institutional animal ethics approval was not required
122 because the study involved invertebrates. All procedures complied with applicable regulations
123 governing the collection and handling of wildlife in New Zealand and followed the Association
124 for the Study of Animal Behaviour/Animal Behavior Society Guidelines for the Treatment of
125 Animals in Behavioural Research and Teaching (“Guidelines for the Ethical Treatment of
126 Nonhuman Animals in Behavioural Research and Teaching,” 2023).

127 **Statistical analyses**

128 All analyses were conducted in R (R Core Team, 2025). I used *observed pairing status* to
129 describe whether an individual was collected as part of a male–female pair (paired = 1) or as a
130 singleton (unpaired = 0). This cross-sectional measure is a component or proxy of mating
131 success rather than a measure of lifetime mating or reproductive success. An association with
132 pairing status could arise through differences in mate encounter, acceptance, mounting success,
133 pair duration, or detectability. The selection analyses therefore assume that phenotypic
134 differences between paired and unpaired individuals contain information about their relative
135 propensity to be observed paired during the sampling period.

136 Sex differences in total body length and total hind leg length were assessed using separate linear
137 models of the form `trait ~ sex`. I also used a multivariate analysis of variance, with body and
138 hind-leg length as the joint response and sex as the predictor, to test for overall sexual
139 dimorphism in the two traits.

140 Phenotypic selection through observed pairing status was analysed separately for males and
141 females. The male analysis included 100 individuals, of which 25 were paired, and 75 were
142 unpaired; the female analysis included 77 individuals, of which 25 were paired, and 52 were
143 unpaired. For each sex, I fitted a generalized additive model using `mgcv::gam()` with the
144 formula

145 `mated ~ s(total_body, bs = "tp", k = 10) + s(total_leg, bs = "tp", k = 10)`.

146 Pairing status was modelled using a binomial error distribution and logit link. Body and hind leg
147 length were represented by separate penalized thin-plate regression splines, each with a basis
148 dimension of (k=10), and smoothing parameters were selected using the `method = "GCV.Cp"`
149 option (Wood, 2017). The models were additive on the logit scale and did not include an explicit
150 tensor-product interaction between the two traits.

151 I estimated standardized selection differentials and gradients following Lande and Arnold (1983)
152 using the *gsg* package (Morrissey & Sakrejda, 2013). The GAMs were fitted to traits measured
153 in millimetres, but *gsg* standardized the resulting differentials and gradients to units of one
154 within-sex phenotypic standard deviation. Standardized directional and quadratic selection
155 differentials were calculated separately for each trait using `moments.differentials()` with
156 10,000 bootstrap replicates. Directional differentials (*S*) describe the total association between a

157 trait and pairing status, whereas quadratic differentials (C) describe changes in trait variance
158 associated with pairing status.

159 Standardized linear, diagonal quadratic, and correlational selection gradients were calculated
160 from each fitted GAM using `gam.gradients()`. Linear gradients (β) describe the average
161 directional association between a trait and relative pairing status after accounting for the other
162 trait. Diagonal quadratic gradients (γ_{ii}) describe average response-scale curvature, whereas the
163 correlational gradient (γ_{ij}) is the response-scale cross-derivative describing how the association
164 between one trait and pairing status varies with the other trait (Brodie et al., 1995; Phillips &
165 Arnold, 1989). Because the models used a nonlinear logit link, the response-scale correlational
166 gradient can be nonzero even though the GAM predictor contained separate additive smooths
167 rather than an explicit interaction term.

168 The `gam.gradients()` function derives gradients from the fitted absolute-fitness function,
169 divides them by mean fitness, and standardizes them by the observed phenotypic standard
170 deviations. The returned estimates are therefore expressed on the standardized relative-fitness
171 scale (Morrissey & Sakrejda, 2013). Diagonal quadratic gradients and their standard errors were
172 reported exactly as returned by `gam.gradients()`; no additional doubling was applied. Standard
173 errors and two-sided P -values were obtained from 10,000 parametric-bootstrap replicates
174 (`se.method = "boot.para"`). During bootstrapping, smoothing parameters were fixed at the
175 values estimated from the original data (`refit.smooth = FALSE`). Values returned as zero were
176 reported as $P < 0.001$. The gradients were estimated separately by sex, so statistical support for a
177 gradient in one sex but not the other does not by itself demonstrate that the corresponding
178 gradients differed between the sexes.

To evaluate whether the apparent contrast in correlational selection was supported by a formal sex comparison, I fitted an additional pooled quadratic logistic regression. Body length and hind leg length were standardized separately within each sex to means of zero and standard deviations of one. Pairing status was modelled as a binary response variable with a binomial error distribution and a logit link. The model included sex; the linear, squared, and cross-product terms for standardized body and hind leg length; and interactions between sex and each morphological term. The sex $\times$ body length $\times$ hind leg length term tested whether the logit-scale body length $\times$ hind leg length association differed between females and males. Because the male and female in an observed pair shared the same pairing event, I calculated cluster-robust standard errors with observed pair as the clustering variable; unpaired individuals were assigned unique singleton clusters. Inference was based on a two-sided cluster-robust Wald test. This pooled test was used to assess sex heterogeneity in the cross-product association and is complementary to, but not numerically identical to, the response-scale correlational gradients returned by `gam.gradients()`. The pooled model was fitted using `glm`, and the clustered covariance matrix and Wald test were calculated using the *sandwich* (Zeileis et al., 2020) and *lmtest* (Zeileis & Hothorn, 2002) packages, respectively.

For the one-dimensional visualizations, I used `fitness.landscape()` to calculate predicted mean pairing probability across 25 values of the focal trait spanning one phenotypic standard deviation below to one phenotypic standard deviation above its sex-specific mean. When body length was the focal trait, hind leg length was retained at its observed values for each individual, and vice versa; predictions were then averaged over the observed distribution of the alternate trait. Thus, these curves are marginal fitness landscapes rather than predictions conditional on a

201 single fixed value of the alternate trait. Shaded regions and dashed lines show the central 95%
202 parametric-bootstrap intervals based on 1,000 replicates, with smoothing parameters held fixed.

203 The two-dimensional male response surface was generated directly from the fitted male GAM.
204 Predicted pairing probabilities were calculated on a 300×300 grid spanning the observed ranges
205 of male body and hind leg length. Predictions for trait combinations outside the convex hull of
206 the observed male phenotypes were excluded to avoid displaying unsupported extrapolations,
207 and the observed male phenotypes were superimposed on the surface.

208 The relationship between female body length and egg load was analysed using a negative-
209 binomial generalized linear model fitted with `MASS::glm.nb()` to accommodate overdispersion
210 in egg counts (Venables & Ripley, 2002). The model formula was `eggs ~ log(total_body)`,
211 with the default logarithmic link. This analysis estimates the association between body length
212 and egg load at collection; I interpret a positive association as consistent with, rather than
213 definitive evidence of, fecundity selection.

214 I assessed body-length assortment among the 25 observed pairs by regressing standardized
215 female body length on standardized male body length (Crespi, 1989; Jiang et al., 2013). I also
216 conducted a two-sided randomization test with 10,000 iterations. In each iteration, females were
217 randomly reassigned among the sampled males, the regression slope was recalculated, and the
218 absolute randomized slope was compared with the absolute observed slope. The randomization
219 *P*-value was the proportion of iterations producing a slope at least as extreme as the observed
220 value.

Results

Female body length and egg load

Female egg load increased with body length (Figure 2). Egg number was positively associated with log-transformed body length ($\beta = 4.50 \pm 1.33$, $z = 3.39$, $p < 0.001$). This association is consistent with fecundity selection favouring greater female body length, while recognizing that egg load is a snapshot of current reproductive state rather than lifetime fecundity.

Phenotypic selection through observed pairing status

The directional selection gradients on body and hind leg length through observed pairing status were qualitatively similar in females and males (Table 1; Figure 3). In both sexes, body length was positively associated with pairing status (males: $\beta_{\text{body}} = 0.97 \pm 0.25$, $p < 0.001$; females: $\beta_{\text{body}} = 0.93 \pm 0.39$, $p = 0.032$), whereas hind leg length was negatively associated with pairing status (males: $\beta_{\text{leg}} = -0.95 \pm 0.25$, $p < 0.001$; females: $\beta_{\text{leg}} = -1.22 \pm 0.33$, $p < 0.001$). The two sexes therefore did not experience opposing directional selection regimes on these traits. These associations may reflect differences in encounter, mounting, attachment duration, or other unmeasured processes and should not be automatically interpreted as differences in lifetime reproductive success.

Despite the qualitative similarity of the directional gradients, the male analysis also supported nonlinear and correlational selection (Table 1; Figure 4). Both male traits had significant positive diagonal quadratic gradients (males: $\gamma_{\text{ii_body}} = 0.60 \pm 0.35$, $p < 0.001$; $\gamma_{\text{ii_leg}} = 0.58 \pm 0.36$, $p < 0.001$) and there was significant negative correlational selection between male body and hind leg length ($\gamma_{\text{ij}} = -0.59 \pm 0.34$, $p < 0.001$). The fitted surface therefore indicated that, on the probability scale, the association between either male trait and observed pairing status varied with the value of the other trait. Within the observed phenotypic range, the predicted pairing

probability was highest among males that combined relatively large body length with short hind legs.

The female analysis provided no statistical support for nonlinear selection on body length ($\gamma_{ii_body} = -0.59 \pm 2.16$, $p=0.391$), nonlinear selection on hind leg length ($\gamma_{ii_leg} = 0.45 \pm 0.41$, $p=0.105$), or correlational selection between the traits ($\gamma_{ij} = -0.23 \pm 0.48$, $p=0.414$). However, the difference in statistical support between the separate male and female analyses did not establish a sex difference. In the pooled quadratic logistic model, the sex $\times$ body length $\times$ hind leg length term provided little evidence that the logit-scale cross-product association differed between the sexes ($z = -1.31$, $P = 0.190$). Thus, the correlational pattern was supported within males, but the data did not clearly demonstrate that it was stronger in males than in females.

Sexual dimorphism

The sampled adults comprised 100 males and 77 females (56.5% male), but this opportunistic sample cannot be used to infer the population sex ratio. Females were larger than males in both total body length and total hind leg length (Figure 5). Mean body length was 9.95 ± 0.07 mm ($n = 77$) in females and 8.95 ± 0.04 mm ($n = 100$) in males, so females were approximately 11.2% longer (female:male ratio = 1.11). Males were, on average, 1.00 mm shorter than females ($\beta = -1.00 \pm 0.08$, $t = -13.34$, $p<0.001$; $R^2 = 0.50$).

Females also had longer hind legs than males (7.05 ± 0.04 mm versus 6.68 ± 0.03 mm), a difference of 5.5% (female:male ratio = 1.06). This difference was statistically supported ($\beta = -0.37 \pm 0.05$, $t = -6.95$, $p<0.001$; $R^2 = 0.22$). The multivariate sex effect was also supported (MANOVA: Pillai's trace = 0.57, $F_{2,174} = 117.46$, $p<0.001$).

Assortative mating

There was no evidence of assortative mating in body length among the 25 observed pairs. The relationship between male and female body length was weak and non-significant ($\beta = -0.23 \pm 0.20$, $t = -1.14$, $p=0.27$), and the observed slope did not differ from the randomized distribution ($P = 0.27$). Thus, body length did not measurably structure which sampled male was paired with which sampled female.

Discussion

Females were larger than males in both body and hind leg length, and female body length was positively associated with egg load. Based on observed pairing status, the directional selection gradients on body and hind leg length were qualitatively similar in females and males: body length was positively associated with pairing, and hind leg length was negatively associated with pairing in both sexes. The two sexes therefore did not experience opposing directional selection regimes on these traits. The male analysis additionally supported nonlinear and correlational selection, with the highest predicted pairing probability among males being for combinations of relatively large bodies and short hind legs, whereas the corresponding female gradients were not statistically supported. Nevertheless, the pooled comparison provided little evidence that the body length $\times$ hind leg length association differed between the sexes. The apparent contrast between the separate analyses therefore should not be interpreted as demonstrating male-specific correlational selection. Body length was not assortative within observed pairs. My findings document female-biased SSD and several contemporary morphology–fitness–component associations, but they do not demonstrate that the measured selection components caused or currently maintain the dimorphism.

The positive association between female body length and egg load is consistent with a common mechanism proposed for female-biased SSD: larger females can allocate more space or resources to egg production (Blanckenhorn, 2005; Honek, 1993; Kingsolver & Pfennig, 2004; Pincheira-Donoso & Hunt, 2017). The inference is nevertheless narrower than a direct estimate of lifetime fecundity selection. The number of eggs present at dissection represents current egg load and may vary with age, reproductive stage, previous oviposition, and the rate at which eggs mature (e.g. Kelly et al., 2024). Consequently, the result shows that longer females carried more eggs at collection; it does not establish how body length affects lifetime egg production, offspring recruitment, or population growth. Repeated measures of oviposition or lifetime reproductive output would provide a stronger test.

The associations between morphology and observed pairing status also require careful interpretation. Paired males were not followed across a mating season, and the survey did not record the number of females encountered, copulations completed, offspring sired, or the duration for which each male had been mounted. A phenotype associated with prolonged attachment could be overrepresented among observed pairs even if it did not increase the number of matings or fertilizations. The male gradients may therefore reflect mate searching, mounting, female acceptance, attachment duration, detectability, or some combination of these processes. Likewise, the field observations are consistent with a scramble-like system because males were not seen defending resources, and pair formation and male–male interactions were not systematically observed. The present data cannot exclude female choice, interference among males, or other forms of competition.

The qualitative similarity of the directional gradients between the sexes is itself a central feature of these results. Both females and males showed positive body-length gradients and negative

hind-leg gradients through pairing status, with similar estimated magnitudes. The supported male correlational gradient adds a second layer: within males, the association between body length and pairing depended on hind-leg length, and the highest predicted pairing probabilities occurred among relatively large-bodied, short-legged males. The biological mechanism behind this combination is unresolved, and several hypotheses—concerning movement, leverage, stability while mounted, or attachment duration—are plausible but not testable with these data. Although the corresponding correlational gradient was not supported in females, the pooled sex comparison provided little evidence that the body length $\times$ hind leg length association differed between females and males. The male-only statistical support may therefore reflect sampling variation or lower precision in the female analysis rather than a genuinely sex-specific selection surface. At the same time, failure to detect an interaction is not evidence that the sexes are equivalent, and the uncertainty around the sex contrast leaves room for biologically meaningful differences. Existing dimorphism may therefore reflect selection acting in earlier generations, sex-specific viability selection, development, ecological differences, genetic architecture, or fitness components not measured here (Blanckenhorn, 2000; Fairbairn, 1997; Fairbairn et al., 2007). The greater proportional dimorphism in body length than in hind leg length also does not map straightforwardly onto the contemporary gradients. This mismatch is informative because it cautions against inferring the evolutionary cause of SSD from a single episode or component of phenotypic selection.

The absence of body-length assortment indicates that, among the 25 pairs sampled, large males were no more likely to be paired with large females than expected by chance. This result does not contradict selection through pairing status. Morphology may affect the probability of being paired while having little effect on the phenotype of the eventual partner (McDonald & Pizzari,

2016). The modest number of pairs nevertheless limits power to detect weak assortment, and a single nocturnal snapshot cannot reveal whether assortment varies through the season or across sites.

The scope of inference is necessarily limited to this population, sampling interval, and taxon. A single-species result cannot establish a general rule for flightless weevils or scramble-competitive arthropods. Its broader contribution is instead comparative: it adds an example in which female-biased SSD co-occurs with a positive body-length–egg-load association, similar directional pairing-status gradients in the sexes, and a male analysis that supported a complex multivariate surface without clear evidence from the pooled comparison that this surface differed between the sexes. Determining whether these features recur in related weevils will require replicated sampling across populations, dates, and species, ideally combined with direct observation of mate searching, pair formation, pair duration, and reproductive success.

Several limitations are therefore central rather than peripheral. The 25 observed pairs came from one site and a restricted 11-night period; pairing status was binary and cross-sectional; egg load was measured once; and behaviour linking morphology to pairing was not recorded directly. The opportunistically collected sex ratio may also reflect sex-specific activity or detectability rather than population abundance. These constraints increase uncertainty about temporal repeatability and the biological processes underlying the statistical associations. Future work should combine repeated field surveys with behavioural observations, experimental measures of locomotion and attachment, and more complete estimates of reproductive success for females and males.

In conclusion, this study documents female-biased SSD, a positive association between female body length and egg load, and qualitatively similar directional gradients in body and hind-leg length, as observed in the pairing status of female and male *Lyperobates* sp. A. The male

analysis supported a dependence of pairing probability on the combination of body and hind-leg length, whereas the corresponding female gradient was not statistically supported. However, the pooled comparison provided little evidence that this association differed between the sexes, so the result does not establish male-specific correlational selection. The cross-sectional measures of egg load and pairing status, together with the qualitative similarity of the directional gradients between the sexes, also prevent a strong causal inference that these contemporary associations produced the observed SSD. The results are best interpreted as evidence that contemporary components of selection in this population are incompletely aligned with existing sexual dimorphism.

366 **Data availability**

367 Data are available through the OSF repository at

368 https://osf.io/khqgs/overview?view_only=728733529d06410ab17086f648a49a38.

369

References

- Andersson, M. B. (1994). *Sexual Selection*. Princeton University Press.
- Blanckenhorn, W. U. (2000). The evolution of body size: What keeps organisms small? *The Quarterly Review of Biology*, 75(4), 385–407.
- Blanckenhorn, W. U. (2005). Behavioral causes and consequences of sexual size dimorphism. *Ethology*, 111(11), 977–1016. <https://doi.org/10.1111/j.1439-0310.2005.01147.x>
- Brodie, E. D., Moore, A. J., & Janzen, F. J. (1995). Visualizing and quantifying natural selection. *Trends Ecol Evol*, 10(8), 313–318. [https://doi.org/10.1016/S0169-5347\(00\)89117-X](https://doi.org/10.1016/S0169-5347(00)89117-X)
- Brown, S. D. J. (2017). Austromonticola, a new genus of broad-nosed weevil (Coleoptera, Curculionidae, Entiminae) from montane areas of New Zealand. *Zookeys*, (707), 73–130. <https://doi.org/10.3897/zookeys.707.12649>
- Crespi, B. J. (1989). Causes of assortative mating in arthropods. *Animal Behaviour*, 38(6), 980–1000. [https://doi.org/10.1016/S0003-3472\(89\)80138-1](https://doi.org/10.1016/S0003-3472(89)80138-1)
- Darwin, C. (1871). *The Descent of Man, and Selection in Relation to Sex* (pp. viii, 423). John Murray. <https://doi.org/10.1037/12293-000>
- Emlen, S. T., & Oring, L. W. (1977). Ecology, sexual selection, and the evolution of mating systems. *Science*, 197(4300), 215–223. <https://doi.org/10.1126/science.327542>
- Fairbairn, D. J. (1997). Allometry for sexual size dimorphism: Pattern and process in the coevolution of body size in males and females. *Annual Review of Ecology, Evolution, and Systematics*, 28(Volume 28, 1997), 659–687. <https://doi.org/10.1146/annurev.ecolsys.28.1.659>
- Fairbairn, D. J., Blanckenhorn, W. U., & Székely, T. (2007). *Sex, Size and Gender Roles*. Oxford University Press.
- Ghiselin, M. T. (1974). *The Economy of Nature and the Evolution of Sex*. Univ of California Press.
- Guidelines for the ethical treatment of nonhuman animals in behavioural research and teaching. (2023). In ASAB. <https://www.asab.org/ethics>; Association for the Study of Animal Behaviour/Animal Behavior Society.
- Haran, J., Li, X., Allio, R., Shin, S., Benoit, L., Oberprieler, R. G., Farrell, B. D., Brown, S. D. J., Leschen, R. A. B., Kergoat, G. J., & McKenna, D. D. (2023). Phylogenomics illuminates the phylogeny of flower weevils (Curculioninae) and reveals ten independent origins of brood-site pollination mutualism in true weevils. *Proc Biol Sci*, 290(2008), 20230889. <https://doi.org/10.1098/rspb.2023.0889>

405 Herberstein, M. E., Painting, C. J., & Holwell, G. I. (2017). Scramble competition polygyny in
 406 terrestrial arthropods. In *Advances in the of* (pp. 237–295). Elsevier.
 407 <https://doi.org/10.1016/bs.asb.2017.01.001>

408 Honek, A. (1993). Intraspecific variation in body size and fecundity in insects: A general
 409 relationship. *Oikos*, 483–492. <https://doi.org/10.2307/3544943>

410 Jiang, Y., Bolnick, D. I., & Kirkpatrick, M. (2013). Assortative mating in animals. *American*
 411 *Naturalist*, 181(6), E125–38. <https://doi.org/10.1086/670160>

412 Kelly, C. D. (2020). Sexual selection on the size and shape of morphological traits in the
 413 Japanese beetle (*Popillia Japonica*). *Behavioral Ecology*, 31, 1073–1083.
 414 <https://doi.org/10.1093/beheco/araa054>

415 Kelly, C. D., Bussière, L. F., & Gwynne, D. T. (2008). Sexual selection for male mobility in a
 416 giant insect with female-biased size dimorphism. *The American Naturalist*, 172(3), 417–
 417 423. <https://doi.org/10.1086/589894>

418 Kelly, C. D., & Gwynne, D. T. (2023). Mating assortment and the strength of sexual selection
 419 in a polyandrous population of Cook Strait giant weta. *Behavioral Ecology*, 34, 506–513.
 420 <https://doi.org/10.1093/beheco/arad017/49802299/arad017.pdf>

421 Kelly, C. D., Peruzzini, W., Chasse-Bilodeau, R., Roche, D. G., & Olivier-Montiglio, P. (2024).
 422 Effect of age and mating on body condition, fecundity and metabolic rate in female sand
 423 crickets,. *Animal Behaviour*, 213, 23–32. <https://doi.org/10.1016/j.anbehav.2024.04.005>

424 Kingsolver, J. G., & Pfennig, D. W. (2004). Individual-level selection as a cause of Cope's rule
 425 of phyletic size increase. *Evolution*, 58(7), 1608–1612. <https://doi.org/10.1111/j.0014-3820.2004.tb01740.x>

427 Lande, R., & Arnold, S. J. (1983). The measurement of selection on correlated characters.
 428 *Evolution*, 37(6), 1210–1226. <https://doi.org/10.2307/2408842>

429 May, B. M. (1993). Larvae of Curculionoidea (Insecta: Coleoptera): A systematic overview.
 430 *Fauna of New Zealand*, 28.

431 McDonald, G. C., & Pizzari, T. (2016). Why patterns of assortative mating are key to study
 432 sexual selection and how to measure them. *Behav Ecol Sociobiol*, 70, 209–220.
 433 <https://doi.org/10.1007/s00265-015-2041-7>

434 Morrissey, M. B., & Sakrejda, K. (2013). Unification of regression-based methods for the
 435 analysis of natural selection. *Evolution*, 67(7), 2094–2100.
 436 <https://doi.org/10.1111/evo.12077>

437 Moya-Laraño, J., Halaj, J., & Wise, D. H. (2002). Climbing to reach females: Romeo should be
 438 small. *Evolution*, 56(2), 420–425.

439 Phillips, P. C., & Arnold, S. J. (1989). Visualizing multivariate selection. *Evolution*, 43(6),
 440 1209–1222. <https://doi.org/10.2307/2409357>

441 Pincheira-Donoso, D., & Hunt, J. (2017). Fecundity selection theory: Concepts and evidence.
 442 *Biol Rev Camb Philos Soc*, 92(1), 341–356.

443 R Core Team. (2025). *R: A language and environment for statistical computing*. R
 444 Foundation for Statistical Computing.

445 Schindelin, J., Arganda-Carreras, I., Frise, E., Kaynig, V., Longair, M., Pietzsch, T., Preibisch, S.,
 446 Rueden, C., Saalfeld, S., Schmid, B., Tinevez, J.-Y., White, D. J., Hartenstein, V., Eliceiri, K.,
 447 Tomancak, P., & Cardona, A. (2012). Fiji: An open-source platform for biological-image
 448 analysis. *Nature Methods*, 9(7), 676–682. <https://doi.org/10.1038/nmeth.2019>

449 Venables, W. N., & Ripley, B. D. (2002). *Modern Applied Statistics with S*. Springer.

450 Wood, S. N. (2017). *Generalized Additive Models: An Introduction with R, Second Edition* (2nd
 451 ed.). Chapman and Hall/CRC. <https://doi.org/10.1201/9781315370279>

452 Zeileis, A., & Hothorn, T. (2002). Diagnostic Checking in Regression Relationships. *R News*,
 453 2, 7–10.

454 Zeileis, A., Köll, S., & Graham, N. (2020). Various versatile variances: An object-oriented
 455 implementation of clustered covariances in R. *Journal of Statistical Software*, 95(1), 1–36.
 456 <https://doi.org/10.18637/jss.v095.i01>

457 Zhu, M., Niklas, K. J., Chen, L., Wang, L., Jiao, Y., & Shi, P. (2025). Scaling relationships and
 458 sexual size dimorphism among the body parts of *Holotrichia Oblita* (Coleoptera:
 459 Scarabaeidae). *Ecol Evol*, 15(7), e71760. <https://doi.org/10.1002/ece3.71760>

460

Table 1: Selection differentials (S, C) and standardized linear (β) and diagonal quadratic (γ) selection gradients through observed pairing status for female and male *Lyperobates* sp. A weevils. Estimates are means $\pm$ SE. Differentials quantify total phenotypic selection, whereas gradients describe associations with each trait after accounting for the other trait. Estimates were returned directly by `gsg::gam.gradients()` on the relative-fitness scale; diagonal quadratic gradients and SEs were not doubled. Standard errors and two-sided *P*-values were obtained from 10,000 parametric-bootstrap replicates.

Trait	Differentials				Gradients			
	$S_i \pm SE$	<i>P</i>	$C_{ii} \pm SE$	<i>P</i>	$\beta_i \pm SE$	<i>P</i>	$\gamma \pm SE$	<i>P</i>
(a) Females								
body length	-0.040 $\pm$ 0.146	0.788	-0.388 $\pm$ 0.220	0.090	0.930 $\pm$ 0.391	0.032	-0.593 $\pm$ 2.163	0.391
hind leg length	-0.148 $\pm$ 0.153	0.335	-0.246 $\pm$ 0.217	0.283	-1.224 $\pm$ 0.332	<0.001	0.454 $\pm$ 0.406	0.105
body length $\times$ hind leg length							-0.228 $\pm$ 0.483	0.414
(b) Males								
body length	0.210 $\pm$ 0.165	0.205	-0.092 $\pm$ 0.199	0.620	0.968 $\pm$ 0.247	<0.001	0.601 $\pm$ 0.354	<0.001
hind leg length	-0.175 $\pm$ 0.165	0.289	-0.121 $\pm$ 0.213	0.553	-0.954 $\pm$ 0.251	<0.001	0.584 $\pm$ 0.360	<0.001
body length $\times$ hind leg length							-0.593 $\pm$ 0.335	<0.001

Figure captions

Figure 1: Mating pair of *Lyperobates* sp. A weevils. The male is mounted dorsally on the female in a typical copulatory position. (a) Lateral (side) view showing the male positioned atop the female's abdomen, with legs grasping the female's body. (b) Frontal view illustrating the alignment of the pair and relative positioning of the male on the female's dorsum.

Figure 2: Relationship between female body length and egg load. Points represent individual females, and egg load is the number of eggs present in the reproductive tract at dissection. Body length is plotted on a log scale. The line shows the fitted mean from the negative-binomial generalized linear model, and the shaded region is the 95% confidence interval. Egg load increased with body length; this cross-sectional measure is not equivalent to lifetime realized fecundity.

Figure 3: Associations between morphology and observed pairing status in males (a–b) and females (c–d). Panels show the predicted probability of being observed paired from binomial generalized additive models as a function of body length (a, c) and hind leg length (b, d), with the alternate trait held constant. Solid lines are model predictions and dashed lines are 95% intervals estimated by parametric bootstrapping. In both sexes, body length had a positive directional gradient and hind leg length had a negative directional gradient. Pairing status is a cross-sectional proxy and may reflect encounter, mounting, or pair duration rather than lifetime reproductive success.

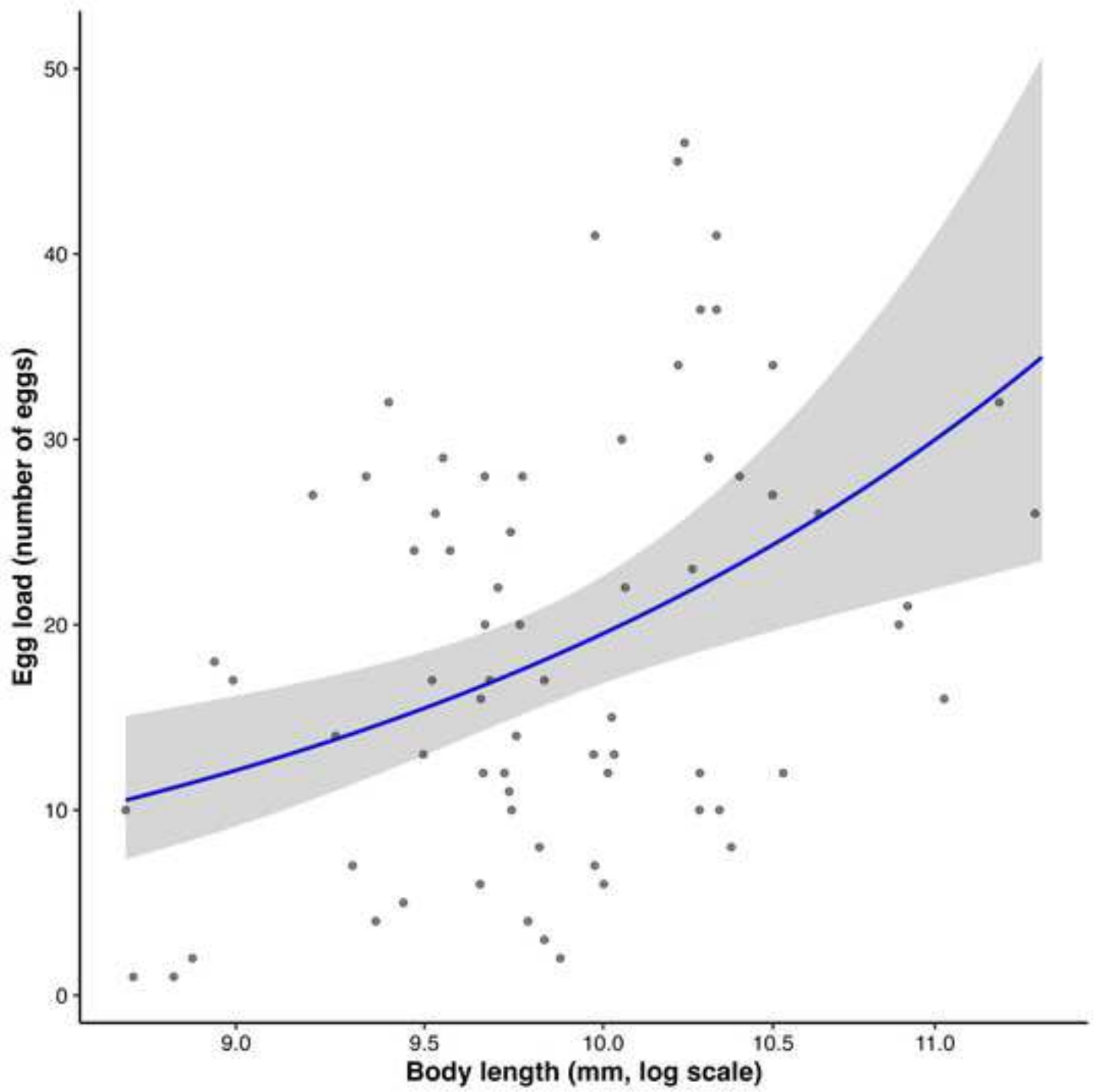
Figure 4: Predicted probability that a male was observed paired as a function of body length and leg length. Colours and labelled contours represent fitted probabilities from the generalized additive model. Points show observed males, and predictions are restricted to the convex hull of the observed combinations of body and leg length. Higher pairing probabilities were associated with greater body length relative to leg length.

Figure 5: Sexual dimorphism in body and hind leg length. Boxplots show total body length and total hind leg length (mm) for females (light grey) and males (dark grey), with points representing individuals. Horizontal lines are medians, boxes are interquartile ranges, and whiskers extend to 1.5× the interquartile range. Females were significantly larger than males for both traits, as assessed by the linear models reported in the Results.



Figure 2

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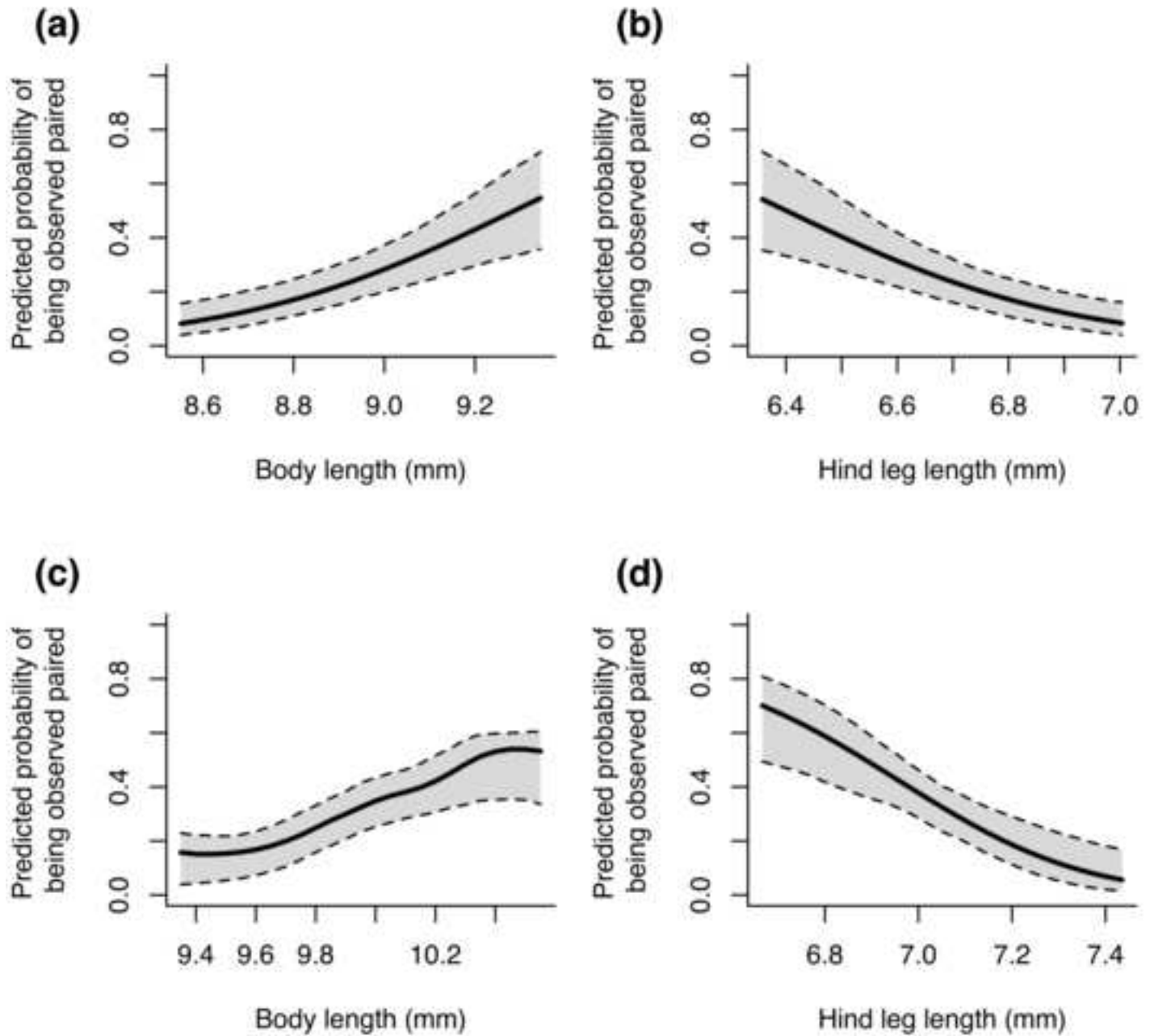


Figure 4

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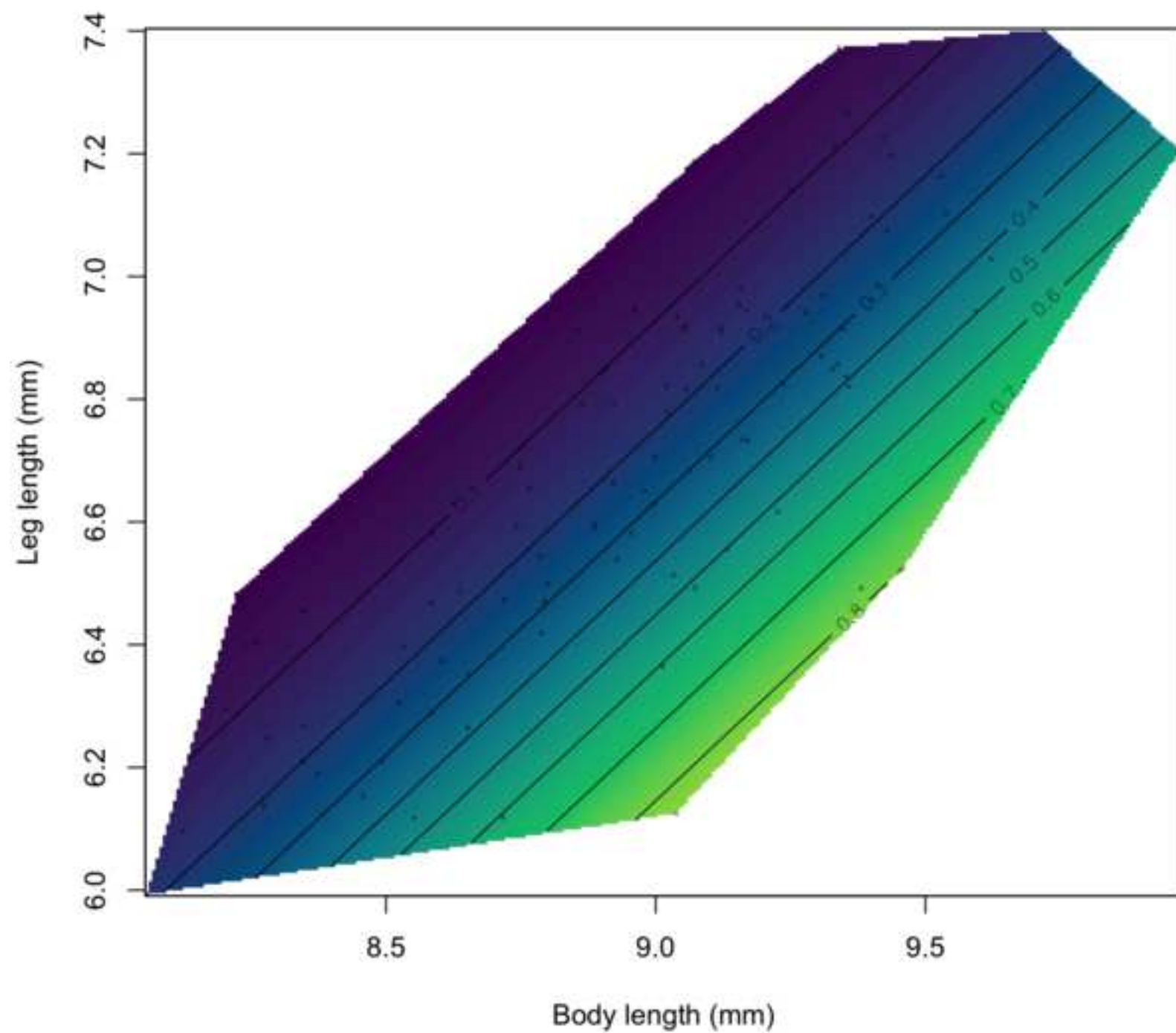


Figure 5

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